From land intelligence to market delivery

An integrated agricultural system for the Amhara Region of Ethiopia that connects land, water, climate, crops, farmers, markets, storage, transport and customers into one coordinated value chain. **Plan backwards from the market to the land; execute forwards from land preparation to the final customer; learn from every season.**



The problem and the approach

Today: push

1. Grow what was grown last year
2. Inputs arrive when they arrive
3. Everyone harvests and sells at once; prices drop
4. Transport is ad hoc; losses are not measured
5. Nobody learns what worked

With the ecosystem: pull

1. Start from what buyers will take
2. Match it to land, monthly water and climate risk
3. Order inputs backwards from planting day
4. Monitor, harvest, store and move on schedule
5. Settle fairly; feed the lessons back

Twelve stages, one learning loop

01 Land Intelligence & Suitability

Where can each crop grow, and how well? The root layer that every later stage builds on.

02 Water Resource Assessment

How much water is actually available for each cluster, in each season, and at what sustainable rate?

03 Climate & Seasonal Intelligence

When does the season start and end, and what weather risks does it carry?

04 Crop Selection & Production Planning

The core of the system: turn buyer demand, suitability, water and climate into a seasonal production plan.

05 Market Intelligence & Demand Forecasting

Start with the market: who buys what, how much, at what quality, when and at what price.

06 Input & Resource Coordination

Make sure seed, fertilizer, equipment and finance arrive before they are needed.

07 Farm Implementation & Production

Carry out the plan in the field with registered farmers, mapped plots and extension support.

08 Monitoring, Forecasting & Risk

Watch crops from space and from the field, forecast yields and raise early warnings.

09 Harvest & Post-harvest Management

Harvest at the right time, keep quality high and make every lot traceable.

10 Storage, Processing & Aggregation

Aggregate smallholder lots into buyer-sized volumes, store safely and add value.

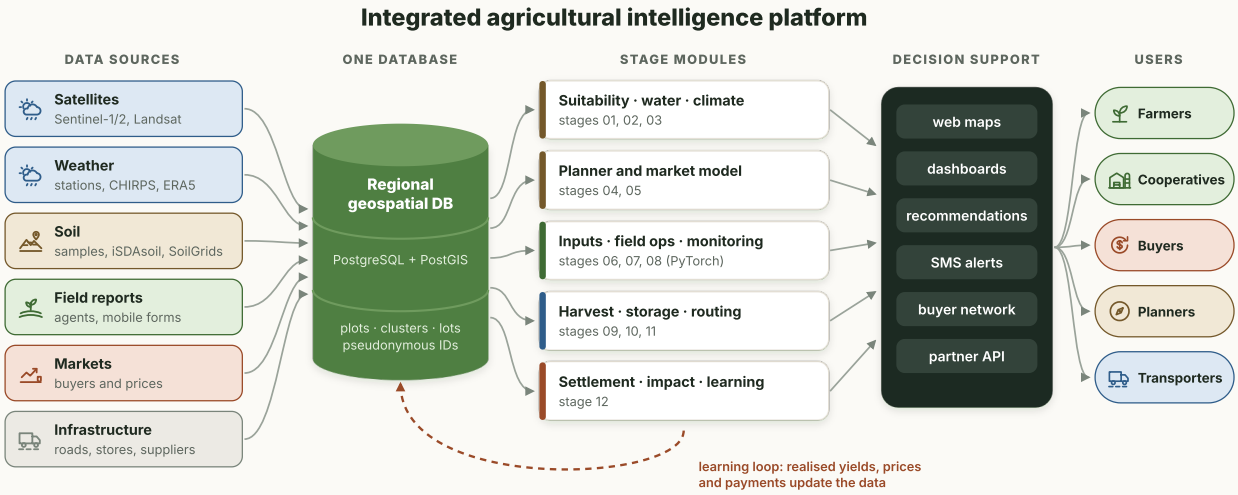
11 Logistics & Transportation

Move produce from farm to collection centre to buyer, on time and at the lowest cost.

12 Market Distribution & Customer Network

Deliver to buyers, settle payments fairly and on time, and feed results back into the next season.

One platform behind the chain



What the stage demos show

Stage	Question	Demo result (synthetic data)
02 Water	Is a seasonal water total enough to plan with?	No: a mix that fits the season total runs short in February; plans use 12 monthly limits.
03 Climate	Does a long-cycle variety fit the season?	A 150-day maize fits only 2 years in 3; early teff fits every year.
04 Planning	Which limits bind, and what is relaxing them worth?	One more m³ of March water: about 50 birr; impossible contracts flagged before planting.
05 Market	Which contract design will both sides honour?	Price band ±15%: honoured 98% of the time versus 77% for a fixed price.
06 Inputs	When must inputs be ordered?	53 days ahead from the union (90% on time); shortfalls flagged in March.
08 Monitoring	Can problems be seen early?	All 6 pest-hit plots warned by day 70; forecast within 1 t of the truth.
09 Harvest	Does harvest order matter?	Deadline-first: zero late plots versus 19; tarpaulins add about 6% value.
10 Storage	Store or sell at harvest?	Hermetic bags: +7% by May; traditional bags never beat selling at harvest.
11 Logistics	How much can routing save?	Shared routes 14% cheaper; rainy-season detours add up to 96%.
12 Distribution	Did incomes really rise?	+8,795 birr per household against comparison kebeles; a naive view would claim +14,722.

Pilot roadmap (18 to 24 months)

MONTHS 1 TO 4
Establish the pilot
- Select one woreda
- Register farmers, map plots
- Assess land, soil, water

MONTHS 3 TO 8
Production and market
- 1 to 2 crops from demand
- Secure inputs
- Buyer agreements before planting

MONTHS 6 TO 14
Deploy the platform
- Geospatial database
- Suitability and water maps
- Advisory and monitoring

MONTHS 12 TO 24
Measure and expand
- Income, yield, losses
- Compare with non-pilot kebeles
- Expand on evidence

Potential partners

Regional and woreda agriculture offices and water authorities; regional universities and agricultural research institutes; national agricultural transformation and research institutes; multilateral food and agriculture organisations and rural development funds; seed and fertilizer suppliers, transporters, warehouses, processors and exporters; farmer cooperatives, unions and water-user associations. These are potential partners to approach; no partnership is implied.